TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	69.4 ± 4.98
Correlation Stability	90.1 ± 0.94
Distribution Consistency	77.8 ± 4.85
Entropy Preservation	63.3 ± 3.80
Mean Z Score Drift	0.3 ± 0.03
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	0.0 ± 0.00
Data Density	100.0 ± 0.00
Coverage Score	98.4 ± 0.66
Feature Accuracy	65.2 ± 3.09
Novelty Score	0.8 ± 0.02

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Dataset Audit: heart_failure_clinical_records.csv

1. Dataset Portrait

Attribute	Value
Rows	299
Columns	13
Numeric features	13
Categorical features	0
Missing cells (%)	0.0%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	google/gemma-4-e2b
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
age	60.834	63.365	+2.531	4.2%	OK
anaemia	0.431	0.339	-0.093	21.4%	Warn
creatinine_phosphokinase	581.839	191.172	-390.668	67.1%	Drift
diabetes	0.418	0.276	-0.142	33.9%	Drift
ejection_fraction	38.084	37.971	-0.113	0.3%	OK
high_blood_pressure	0.351	0.172	-0.180	51.1%	Drift
platelets	263358.029	254556.820	-8801.209	3.3%	OK
serum_creatinine	1.394	1.046	-0.348	25.0%	Warn
serum_sodium	136.625	137.310	+0.684	0.5%	OK
sex	0.649	0.812	+0.163	25.1%	Drift
smoking	0.321	0.038	-0.283	88.3%	Drift
time	130.261	163.439	+33.178	25.5%	Drift

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	69.4 ± 4.98
Correlation Stability (*2)	90.1 ± 0.94
Distribution Consistency (*3)	77.8 ± 4.85
Entropy Preservation (*4)	63.3 ± 3.80
Mean Z Score Drift (*5)	0.3 ± 0.03
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	0.0 ± 0.00

Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	98.4 ± 0.66
Feature Accuracy (*10)	65.2 ± 3.09
Novelty Score (*11)	0.8 ± 0.02

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8066666666666666	0.7116666666666667	0.7783333333333333	-0.02833333333333332	
F1	0.6937863401621888	0.3901399868247693 6	0.6012881701385968	-0.09249817002359192	*

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7883333333333333	0.7233333333333333	0.7716666666666666	-0.01666666666666663	
F1	0.675837074942099	0.5808730799624362	0.6381922570640175	-0.03764481787808155	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.835	0.75	0.8366666666666667	0.00166666666666683	
F1	0.7547418227770806	0.5306693394178759	0.752256924528976	-0.00248489824810466 1	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8316666666666667	0.7433333333333333	0.8283333333333334	-0.00333333333333334 35	
F1	0.7556715392060958	0.547697002410822	0.7396823313140702	-0.01598920789202553 5	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.825	0.755	0.8233333333333335	-0.00166666666666667 18	
F1	0.7417404771442427	0.5253418653418654	0.7308240754611217	-0.01091640168312096 9	

Model: KNN

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Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6133333333333335	0.6333333333333334	0.6333333333333334	0.019999999999999983	
F1	0.007142857142857143	0.016	0.045059288537549404	0.037916431394692265	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7499999999999999	0.7150000000000001	0.7066666666666667	-0.04333333333333337	
F1	0.5667171896628354	0.4349570738266156	0.4646967158182267	-0.10202047384460858	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5316666666666666	0.4933333333333333	0.5533333333333333	0.02166666666666665	
F1	0.27138048447451885	0.20428087908444112	0.24817975382490937	-0.02320073064960949	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6333333333333334	0.6333333333333334	0.6333333333333334	0	
F1	N/A	0.0	N/A	0	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8366666666666667	0.7433333333333334	0.8183333333333334	-0.018333333333333358	
F1	0.7454897762794181	0.4937121948143817	0.7059466952272475	-0.039543081052170706	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	75.0	0.0	582.0	0.0	20.0	1.0	265000.0	1.9	130.0	1.0
2	55.0	0.0	7861.0	0.0	38.0	0.0	263358.03	1.1	136.0	1.0
3	65.0	0.0	146.0	0.0	20.0	0.0	162000.0	1.3	129.0	1.0
4	50.0	1.0	111.0	0.0	20.0	0.0	210000.0	1.9	137.0	1.0
5	65.0	1.0	160.0	1.0	20.0	0.0	327000.0	2.7	116.0	0.0
6	90.0	1.0	47.0	0.0	40.0	1.0	204000.0	2.1	132.0	1.0
7	75.0	1.0	246.0	0.0	15.0	0.0	127000.0	1.2	137.0	1.0
8	60.0	1.0	315.0	1.0	60.0	0.0	454000.0	1.1	131.0	1.0
9	65.0	0.0	157.0	0.0	65.0	0.0	263358.03	1.5	138.0	0.0
10	80.0	1.0	123.0	0.0	35.0	1.0	388000.0	9.4	133.0	1.0
11	75.0	1.0	81.0	0.0	38.0	1.0	368000.0	4.0	131.0	1.0
12	62.0	0.0	231.0	0.0	25.0	1.0	253000.0	0.9	140.0	1.0
13	45.0	1.0	981.0	0.0	30.0	0.0	136000.0	1.1	137.0	1.0
14	50.0	1.0	168.0	0.0	38.0	1.0	276000.0	1.1	137.0	1.0
15	49.0	1.0	80.0	0.0	30.0	1.0	427000.0	1.0	138.0	0.0
16	82.0	1.0	379.0	0.0	50.0	0.0	47000.0	1.3	136.0	1.0
17	87.0	1.0	149.0	0.0	38.0	0.0	262000.0	0.9	140.0	1.0
18	45.0	0.0	582.0	0.0	14.0	0.0	166000.0	0.8	127.0	1.0
19	70.0	1.0	125.0	0.0	25.0	1.0	237000.0	1.0	140.0	0.0
20	48.0	1.0	582.0	1.0	55.0	0.0	87000.0	1.9	121.0	0.0

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B.1 Audit Trace: Fold 1 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	62.859982544284094	0.0	222.0	0.0	12.0	0.0	145941.0	0.9571489765270995	140.0	1.0
2	52.80822005526509	1.0	117.0	1.0	26.0	1.0	236403.0	0.8890809679612339	139.0	0.0
3	55.67634820536468	0.0	158.0	0.0	28.0	0.0	141031.0	1.1566407739656108	136.0	1.0
4	62.19099233901867	0.0	219.0	0.0	34.0	0.0	228194.0	0.8897028997064306	135.0	1.0
5	62.49363709959436	0.0	124.0	0.0	40.0	0.0	297524.0	0.7710460075487622	139.0	1.0
6	64.3933318602094	0.0	134.0	1.0	22.0	0.0	351805.0	0.9022511549537533	136.0	1.0
7	45.0	1.0	66.0	1.0	25.0	0.0	233000.0	0.8	135.0	1.0
8	70.0	0.0	161.0	0.0	25.0	0.0	244000.0	1.2	142.0	0.0
9	58.0	1.0	400.0	0.0	40.0	0.0	164000.0	1.0	139.0	0.0
10	55.0	0.0	572.0	1.0	35.0	0.0	231000.0	0.8	143.0	0.0

B.2 Audit Trace: Fold 2 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	49.715134866456344	0.0	198.0	0.0	41.0	0.0	164061.0	1.0379562952358081	135.0	1.0
2	64.373423453068244	0.0	234.0	0.0	34.0	0.0	220522.0	0.7234254978583747	136.0	1.0
3	55.047036762328474	0.0	29.0	0.0	37.0	0.0	134688.0	0.9347121984117547	140.0	1.0
4	41.732444084795311	0.0	325.0	0.0	31.0	0.0	451000.0	0.7995908259317095	137.0	1.0
5	48.639948937030211	1.0	200.0	0.0	24.0	0.0	320356.0	1.0167835618035426	136.0	1.0
6	54.486647635223111	0.0	242.0	0.0	39.0	0.0	355227.0	1.3700461326111646	140.0	1.0
7	60.0	0.0	2656.0	1.0	30.0	0.0	305000.0	2.3	137.0	1.0
8	45.0	1.0	130.0	0.0	35.0	0.0	174000.0	0.8	139.0	1.0
9	45.0	0.0	582.0	1.0	55.0	0.0	543000.0	1.0	132.0	0.0
10	65.0	1.0	59.0	1.0	60.0	0.0	172000.0	0.9	137.0	0.0

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B.3 Audit Trace: Fold 3 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	48.628681713791075	0.0	196.0	0.0	45.0	0.0	281482.0	0.8789977974641894	139.0	1.0
2	69.15303796534764	0.0	253.0	0.0	26.0	0.0	156915.0	1.2509124263780622	138.0	1.0
3	65.4761991086699	0.0	62.0	0.0	37.0	0.0	291487.0	0.8038165074274695	140.0	1.0
4	67.75563060569156	1.0	271.0	0.0	37.0	0.0	279290.0	0.9018601008316104	141.0	1.0
5	65.31236492026586	1.0	196.0	0.0	32.0	0.0	292723.0	1.138070646870979	130.0	1.0
6	47.952081152791955	0.0	765.0	0.0	34.0	0.0	191408.0	0.9294757409685358	142.0	1.0
7	50.0	0.0	582.0	1.0	38.0	0.0	310000.0	1.9	135.0	1.0
8	79.0	1.0	55.0	0.0	50.0	1.0	172000.0	1.8	133.0	1.0
9	65.0	1.0	160.0	1.0	20.0	0.0	327000.0	2.7	116.0	0.0
10	70.0	0.0	92.0	0.0	60.0	1.0	317000.0	0.8	140.0	0.0

B.4 Audit Trace: Fold 4 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	70.63792206013885	0.0	300.0	0.0	29.0	0.0	213017.0	0.9370306939906867	134.0	1.0
2	77.13640380236718	0.0	286.0	0.0	34.0	0.0	306494.0	1.3166308610765243	141.0	1.0
3	72.1273385715185	0.0	95.0	0.0	33.0	0.0	265029.0	1.2316547164600717	139.0	1.0
4	60.153742520619446	1.0	186.0	0.0	26.0	1.0	239612.0	0.7120370914841233	137.0	0.0
5	69.28927230715333	0.0	201.0	0.0	38.0	0.0	182730.0	0.8721805572093032	139.0	1.0
6	57.218282997504446	1.0	103.0	0.0	40.0	0.0	314339.0	1.0949924698199667	148.0	1.0
7	48.0	1.0	131.0	1.0	30.0	1.0	244000.0	1.6	130.0	0.0
8	95.0	1.0	371.0	0.0	30.0	0.0	461000.0	2.0	132.0	1.0
9	45.0	1.0	981.0	0.0	30.0	0.0	136000.0	1.1	137.0	1.0
10	50.0	0.0	582.0	1.0	38.0	0.0	310000.0	1.9	135.0	1.0

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B.5 Audit Trace: Fold 5 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	56.7366824225543 1	0.0	65.0	0.0	42.0	0.0	277614.0	1.10204963732986 5	139.0	0.0
2	73.3674860569028 2	0.0	152.0	0.0	37.0	0.0	229857.0	1.12212074526268 96	132.0	1.0
3	52.0597431180649 7	0.0	219.0	1.0	38.0	0.0	284538.0	1.06150981579857 14	139.0	1.0
4	56.8231788445204 84	0.0	199.0	0.0	33.0	0.0	248867.0	0.96267133813752 99	136.0	1.0
5	61.6384080066433 4	1.0	111.0	0.0	58.0	0.0	281940.0	0.86533642387941 39	132.0	0.0
6	80.5572879068848 8	0.0	165.0	1.0	29.0	0.0	256884.0	1.04052836935296 3	133.0	1.0
7	78.0	0.0	224.0	0.0	50.0	0.0	481000.0	1.4	138.0	1.0
8	70.0	0.0	582.0	0.0	20.0	1.0	263358.03	1.83	134.0	1.0
9	61.0	0.0	248.0	0.0	30.0	1.0	267000.0	0.7	136.0	1.0
10	40.0	0.0	478.0	1.0	30.0	0.0	303000.0	0.9	136.0	1.0

B.6 Audit Trace: Fold 6 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	56.3215247395014 5	0.0	87.0	0.0	41.0	0.0	206034.0	1.32729868938971 63	134.0	1.0
2	56.0751996953196 25	0.0	83.0	0.0	26.0	1.0	235765.0	1.05668037707568 47	136.0	0.0
3	48.0825470398146 7	0.0	165.0	0.0	65.0	0.0	290365.0	0.73752193601941 81	138.0	1.0
4	74.6442053803757 4	0.0	48.0	0.0	30.0	0.0	261825.0	0.79923519389583 14	139.0	1.0
5	59.6973607742547 1	0.0	105.0	0.0	42.0	0.0	180102.0	0.90770846309304 86	140.0	1.0
6	69.0937702653246 2	1.0	10.0	0.0	33.0	0.0	196024.0	0.94579152449901 56	139.0	1.0
7	81.0	0.0	4540.0	0.0	35.0	0.0	231000.0	1.18	137.0	1.0
8	70.0	0.0	2695.0	1.0	40.0	0.0	241000.0	1.0	137.0	1.0
9	70.0	0.0	835.0	0.0	35.0	1.0	305000.0	0.8	133.0	0.0
10	80.0	0.0	776.0	1.0	38.0	1.0	192000.0	1.3	135.0	0.0

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B.7 Audit Trace: Fold 7 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	56.704512797518916	0.0	214.0	0.0	41.0	0.0	230367.0	0.7177098394469534	136.0	1.0
2	54.566768638113544	0.0	695.0	0.0	21.0	0.0	284648.0	1.2489273190937607	132.0	1.0
3	63.298341704152203	1.0	249.0	0.0	31.0	1.0	336908.0	0.9418642482336906	141.0	1.0
4	69.753539881283509	1.0	23.0	0.0	48.0	0.0	315587.0	1.0784230174558687	137.0	1.0
5	46.5698346099033	0.0	259.0	0.0	32.0	0.0	189792.0	1.3710316159212945	138.0	0.0
6	56.052428104345804	0.0	197.0	0.0	41.0	0.0	434701.0	1.2151852409417914	135.0	1.0
7	75.0	0.0	582.0	1.0	30.0	1.0	263358.03	1.83	134.0	0.0
8	53.0	0.0	207.0	1.0	40.0	0.0	223000.0	1.2	130.0	0.0
9	60.0	1.0	2281.0	1.0	40.0	0.0	283000.0	1.0	141.0	0.0
10	59.0	1.0	280.0	1.0	25.0	1.0	302000.0	1.0	141.0	0.0

B.8 Audit Trace: Fold 8 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	55.852652489938504	0.0	118.0	0.0	22.0	0.0	317091.0	1.0925465376809544	135.0	1.0
2	67.4285201749408	1.0	-35.0	0.0	24.0	0.0	342060.0	1.2553873335326595	140.0	1.0
3	55.448602510650915	0.0	52.0	1.0	31.0	1.0	334199.0	0.9705934608070019	137.0	0.0
4	59.327824132554511	0.0	279.0	0.0	39.0	0.0	307651.0	1.2699426265130636	137.0	1.0
5	81.393231022380009	0.0	258.0	0.0	38.0	0.0	246724.0	1.2551396280528116	135.0	1.0
6	69.8762319824724	0.0	123.0	0.0	58.0	0.0	262096.0	1.0624325487428776	137.0	1.0
7	65.0	0.0	224.0	1.0	50.0	0.0	149000.0	1.3	137.0	1.0
8	80.0	0.0	898.0	0.0	25.0	0.0	149000.0	1.1	144.0	1.0
9	66.0	1.0	72.0	0.0	40.0	1.0	242000.0	1.2	134.0	1.0
10	75.0	1.0	582.0	0.0	30.0	0.0	225000.0	1.83	134.0	1.0

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B.9 Audit Trace: Fold 9 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	58.5307447932723	0.0	75.0	1.0	23.0	0.0	251970.0	1.3015430252397646	139.0	1.0
2	66.8095948190849	0.0	607.0	0.0	39.0	0.0	218999.0	1.3060419817994287	139.0	1.0
3	54.827060690290004	0.0	197.0	0.0	29.0	0.0	206348.0	0.5654969276413488	139.0	1.0
4	50.59846008731023	0.0	520.0	1.0	29.0	0.0	231986.0	1.1060103199317273	143.0	1.0
5	58.953601415930976	0.0	182.0	1.0	21.0	0.0	299992.0	1.2254498900047948	135.0	1.0
6	70.635493791229001	1.0	307.0	0.0	32.0	1.0	211827.0	0.6197405063387407	138.0	1.0
7	55.0	0.0	572.0	1.0	35.0	0.0	231000.0	0.8	143.0	0.0
8	60.0	0.0	68.0	0.0	20.0	0.0	119000.0	2.9	127.0	1.0
9	70.0	0.0	97.0	0.0	60.0	1.0	220000.0	0.9	138.0	1.0
10	53.0	1.0	1808.0	0.0	60.0	1.0	249000.0	0.7	138.0	1.0

B.10 Audit Trace: Fold 10 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	63.171668470411177	0.0	250.0	0.0	36.0	0.0	293813.0	1.1992865801170893	135.0	1.0
2	66.918241397237004	1.0	203.0	0.0	50.0	0.0	280933.0	1.1647858163875873	134.0	1.0
3	65.624693919361709	1.0	66.0	0.0	39.0	0.0	202920.0	1.0365907693689702	144.0	1.0
4	46.397330478987	0.0	87.0	0.0	28.0	0.0	257920.0	0.9705273684077791	139.0	1.0
5	76.940154930138001	0.0	166.0	0.0	29.0	0.0	288498.0	1.2549437805465846	135.0	1.0
6	61.003689866560603	0.0	177.0	1.0	26.0	0.0	262426.0	1.1187590257671347	132.0	1.0
7	60.0	0.0	897.0	1.0	45.0	0.0	297000.0	1.0	133.0	1.0
8	75.0	0.0	582.0	0.0	20.0	1.0	265000.0	1.9	130.0	1.0
9	45.0	1.0	1876.0	1.0	35.0	0.0	226000.0	0.9	138.0	1.0
10	75.0	1.0	246.0	0.0	15.0	0.0	127000.0	1.2	137.0	1.0